

Schur's Theorem

And Fermat's Last Theorem in Finite Fields

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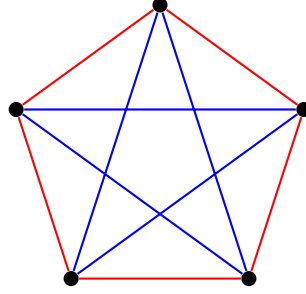
1 Ramsey Number

Let K_n be the complete graph on n vertices.

Definition 1 (Ramsey Number). $R(r_1, \dots, r_k)$ is the smallest number R such that for any k -coloring of K_R , there exists a color i such that K_R contains at least one subgraph K_{r_i} of color i .

Example 1. It is known that $R(3, 3) = 6$. Therefore, for any 2-coloring of the edges of K_6 , one can always find a monochromatic subgraph K_3 .

The following 2-coloring of K_5 shows that $R(3, 3) > 5$: for this coloring, one cannot find any monochromatic K_3 .



Ramsey's theorem guarantees that $R(r_1, \dots, r_k)$ exists and is well-defined.

2 Schur's Theorem

Theorem 1 (Schur). *Given $k > 1$, there exists an integer N such that for all $n \geq N$ and every k -coloring of $\{1, \dots, n\}$, there exist $x, y, z \in \{1, \dots, n\}$ such that $x + y = z$ and x, y, z are of the same color.*

Proof. Consider $N = R(\underbrace{3, \dots, 3}_{k \text{ times}})$. By the definition of the Ramsey number, for any k -coloring C of the edges of K_N and any $n \geq N$, there exists a monochromatic subgraph K_3 .

Now, given any k -coloring v of $\{1, \dots, n\}$, define a k -coloring of the edges of K_n by $C(i, j) = v(j - i)$ for edge (i, j) .

A monochromatic K_3 subgraph of K_n means there exist three numbers a, b, c such that the edges of the three vertices, $C(a, b), C(b, c), C(a, c)$, are of the same color. This implies $v(b - a) = v(c - b) = v(c - a)$.

Let $x = b - a, y = c - b, z = c - a$. Then x, y, z satisfy $x + y = z$ and $v(x) = v(y) = v(z)$. □

We call the smallest such N in Theorem 1 the *Schur number* $s(k)$.

3 Fermat's Last Theorem in Finite Fields

Theorem 2. *Given $n > 1$, there exists a prime p_0 such that for all primes $p \geq p_0$, the equation*

$$x^n + y^n = z^n \pmod{p}$$

has a non-trivial integer solution ($p \nmid xyz$).

Proof. Let $p_0 > s(n)$ be a prime and let $p \geq p_0$ be another prime. Since $p - 1 \geq s(n)$, for any n -coloring of $\{1, \dots, p - 1\}$, there exists a solution (x, y, z) with $x + y = z$ and x, y, z of the same color. Notice that, for any $k < n$, the same conclusion holds for any k -coloring of $\{1, \dots, p - 1\}$.

Consider the group $\mathbb{Z}_p^*$ and its subgroup

$$A_n = \{x^n : x \in \mathbb{Z}_p^*\}.$$

Claim. $|A_n| = \frac{p-1}{\gcd(n, p-1)} =: k.$

Proof of Claim. There are $\gcd(n, p-1)$ solutions ϕ_i of $X^n = 1$. Given $x \in \mathbb{Z}_p^*$, the set $\{x\phi_i\}_i$ consists of all the (disjoint) solutions to $X^n = x^n$. Thus each value x^n is attained by exactly $\gcd(n, p-1)$ elements of $\mathbb{Z}_p^*$, so there are $\frac{p-1}{\gcd(n, p-1)}$ possible values of x^n . □

Consider all cosets of A_n : numbers $a_1, \dots, a_k \in \mathbb{Z}_p^*$ such that the $a_i A_n$ are disjoint and $\mathbb{Z}_p^* = \bigcup_{i \in [k]} a_i A_n$. Define the mapping $v(x) = i$ if $x \in a_i A_n$.

Then v is a k -coloring of $\{1, \dots, p-1\}$. Since $k < n$, there exist x, y, z such that $x + y = z$ and they are of the same color, say color i . This means there exist $x_i, y_i, z_i \in \mathbb{Z}_p^*$ such that

$$x + y = z \quad \wedge \quad x = a_i x_i^n \bmod p, \quad y = a_i y_i^n \bmod p, \quad z = a_i z_i^n \bmod p.$$

Hence

$$a_i x_i^n + a_i y_i^n = a_i z_i^n \bmod p \implies x_i^n + y_i^n = z_i^n \bmod p.$$

□